

TASK 3: Protein Structure Prediction

Title

Prediction and Structural Analysis of Human Insulin Protein Using SWISS-MODEL and AlphaFold Database

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Domain: Bioinformatics

Aim: To predict and analyze the three-dimensional (3D) structure of the Human Insulin protein using bioinformatics tools such as **UniProt**, **SWISS-MODEL**, and **PyMOL**.

Objective

- 1. Retrieve the amino acid sequence from UniProt.
- 2. Obtain the predicted 3D protein structure.
- 3. Visualize the structure using PyMOL.
- 4. Study important structural features of the protein.

Protein Selected

Property	Details
Protein Name	Human Insulin
Organism	Homo sapiens
UniProt ID	P01308
Length	110 amino acids (preproinsulin)
Function	Regulates blood glucose levels

Amino Acid Sequence

MALWMRLPLLLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRGI
VEQCCTSICSLYQLENYCN

P01308 · INS_HUMAN | INS Tools ▾ Download Add Community curated (1) Add a publication Entry feedback

Entry **Variant viewer** **180** **Feature viewer** **Genomic coordinates** **Publications** **External links** **History**

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name 1 **See also** sequence in **UniParc** or sequence clusters in **UniRef**

Tools ▾ Download Add Highlight ▾ Copy sequence

Length 110 **Last updated** 1986-07-21 v1
Mass (Da) 11,981 **MD5 Checksum**ⁱ 12E9C9E4E2835C302E8BA615115EDDA3

MALWMRLRLPL LALLALWGPD PAAAFVNQHL CGSHLVEALY LVCGERGFY TPKTREAED LQVGQVELGG GPGAGSLQPL ALEGSQKRG IVEQCCTSLIC SLYQLENYCN

Tools Used

1. UniProt
2. SWISS-MODEL
3. PyMOL

Procedure

Step 1

Open UniProt.
 Search: Human Insulin P01308
 Download the FASTA sequence.

Step 2

Open SWISS-MODEL Workspace.
 Paste the FASTA sequence.
 Run the prediction.
 Download the PDB model.

SWISS-MODEL Modelling Repository Tools Documenta

Start a New Modelling Project

Target Sequence(s): Target MALWMRLRLPLALLALWGPDPAFAAFVNQHLGSHLVEALYLVCGERGFYTPKTRGIVEQCCTSLICSLYQLENYCN 76
(Format must be FASTA, Clustal, plain string, or a valid UniProtKB AC)

Add Hetero Target Reset

Project Title:

Email:

Search For Templates Build Model

By using the SWISS-MODEL server, you agree to comply with the following [terms of use](#) and to cite the corresponding [articles](#).

Step 3

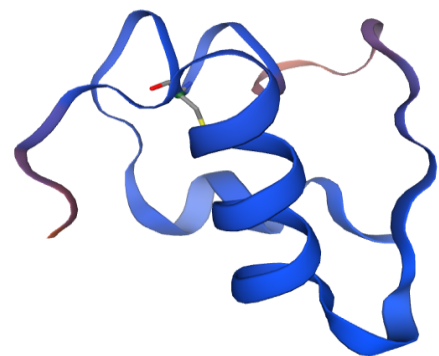
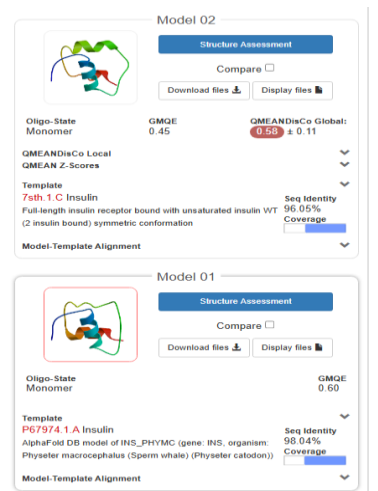
Open PyMOL.
 Load
 insulin.pdb
 Use

Show → Cartoon
Color
Spectrum
Rotate the protein.
Save image.

Results

Predicted Structure

The SWISS- model successfully predicted the tertiary structure of Human Insulin.



Structural Features

Feature	Observation
Protein Type	Hormone
Organism	Human
Alpha Helices	Present
Beta Sheets	Present
Loops	Present
Disulfide Bonds	Three
Confidence	High

